Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To reduce the concern of Interior part scratches-Door trim pad/Rocker panel/Front console/IP by 80% from Scorpio / Marazzo model"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	**10.07.19**	**10.07.19**
Completion Date	**20.11.19**	**27.11.19**

Current Status: **CLOSED**

Cross Functional Team

No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Ganesh Bendale	L7-0	STCF	9850952219
2	**Mentor**	Chikhale Sandip	L5-M	STCF	9922408138
3	**Member 1**	Sachin Bhamare	L7-O	MQA	9158960111
4	**Member 2**	Nandy Tarit	L7-O	PRD	8975767608
5	**Member 3**	Abhishek Sonawane	L8-0	STCF	9881751802
6	**Member 4**	Santosh Jadhav	L7-0	MQA	9423309400
7	**Member 5**	AVINASH MAHALE	L7-0	SCM	9823996005

Step1. Select Subject

Brief description of Problem:-

To reduce the concern of Interior part scratches-Door trim pad/Rocker panel/Front console/IP by 80% from Scorpio / Marazzo model.

What : Scratches on Interior parts.

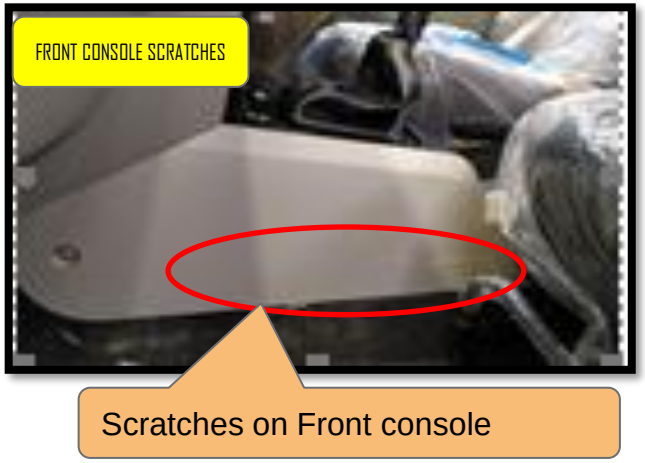
Where : At Trim-3 Buyoff/CAI/RFI

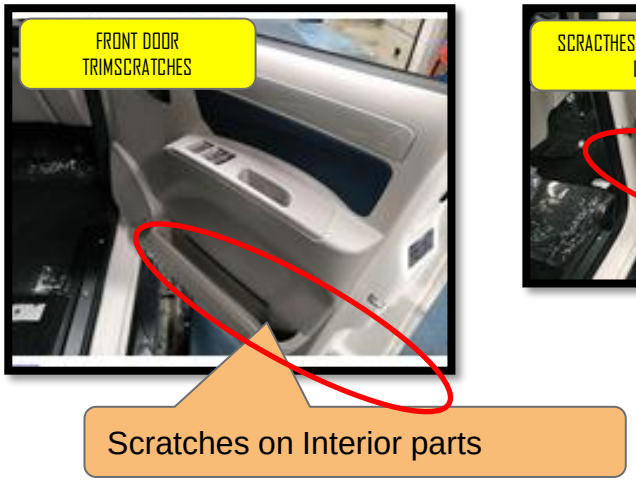
When : Since JAN-19 month

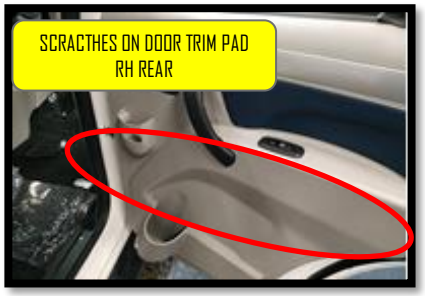
Who : By associate at respective part fitment stage,Buy off check man & RFI inspector.

WHY : High rework, FRC /FBO loss, Repeat issue at Buyoff,CAI & RFI Audit.

HOW : Avg. 51 cases/month reported at Buyoff and 3-4 cases at CAI/RFI.





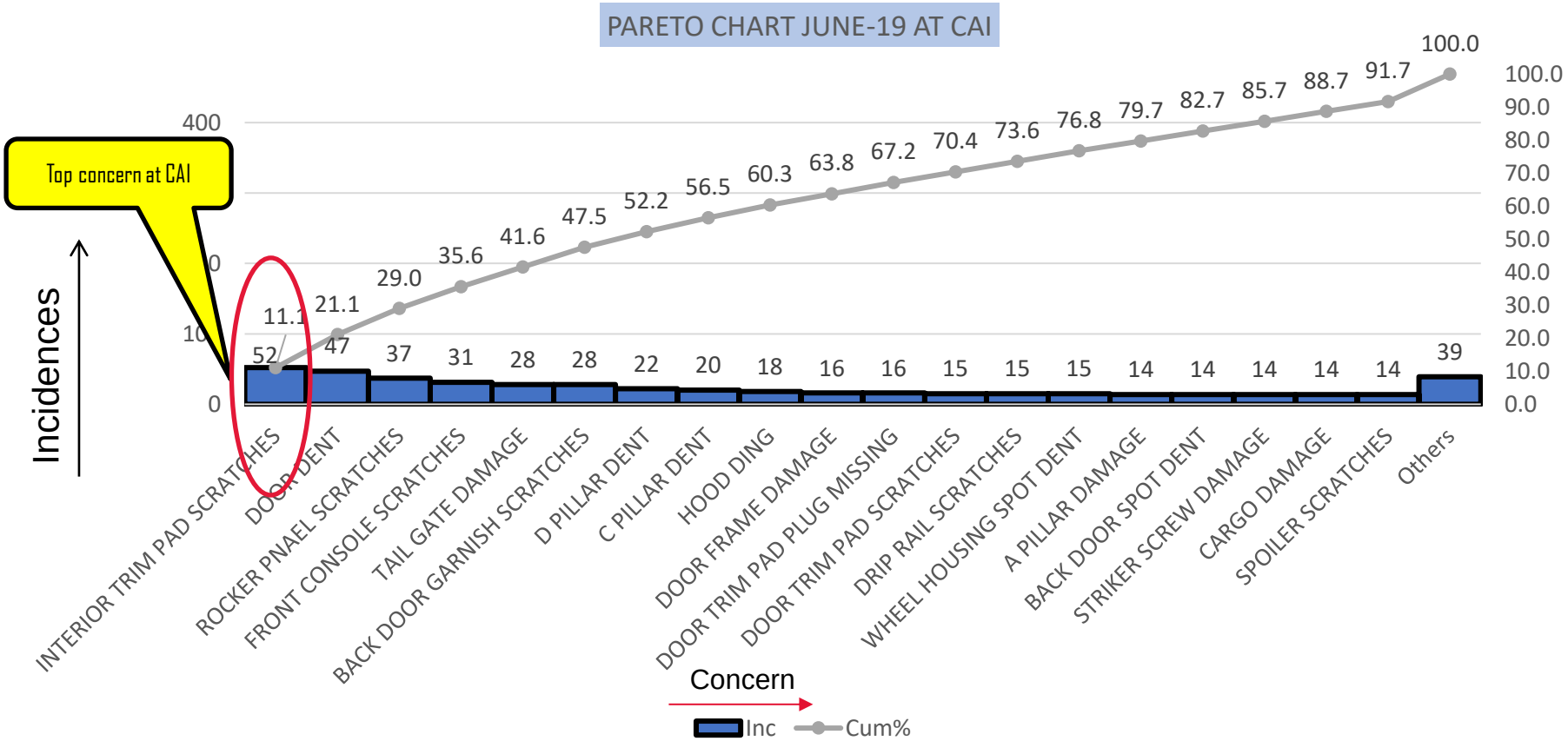


2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
- DPR Improvement
- Offline Rework reduction

2. Justify the choice



Inference:
Interior part scratches is repeated top contributing concern , hence taken for resolution

2. Justify the choice

Severity & Consequences of Concern :-

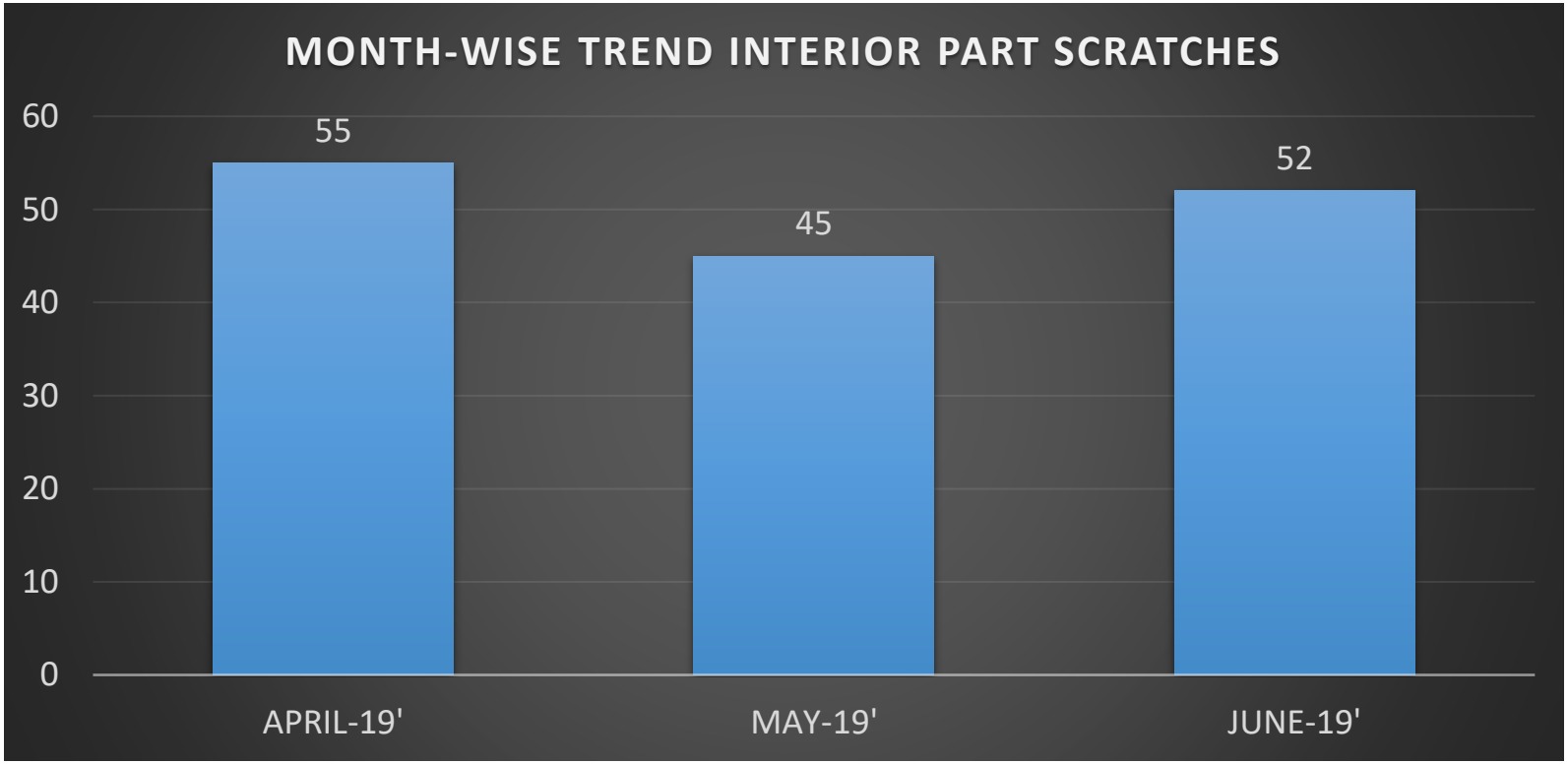
- **Avg. 51 Cases reported per month, for rework, 3-4 V2 concern reported in RFI/GNOAVC audit.**
- **Customer Dissatisfaction due to Bad aesthetics.**
- **Incorrect appearance when customer enters or seat inside veh.**
- **Rework Difficult**
- **Rework cost more.**
- **Rework time 8 min (approx.) for each vehicle by 1 operator.**
- **Rework cost is 76412 Rs/year**

Cost of Poor Quality - Rework

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	4.0
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	254.71
Avg no. of working days	days	25
Operator cost /month	Rs	6367.7
Operator Cost/year	**Rs**	**76412.9**

2. Justify the choice

MONTHWISE DEFECT TREAD :-



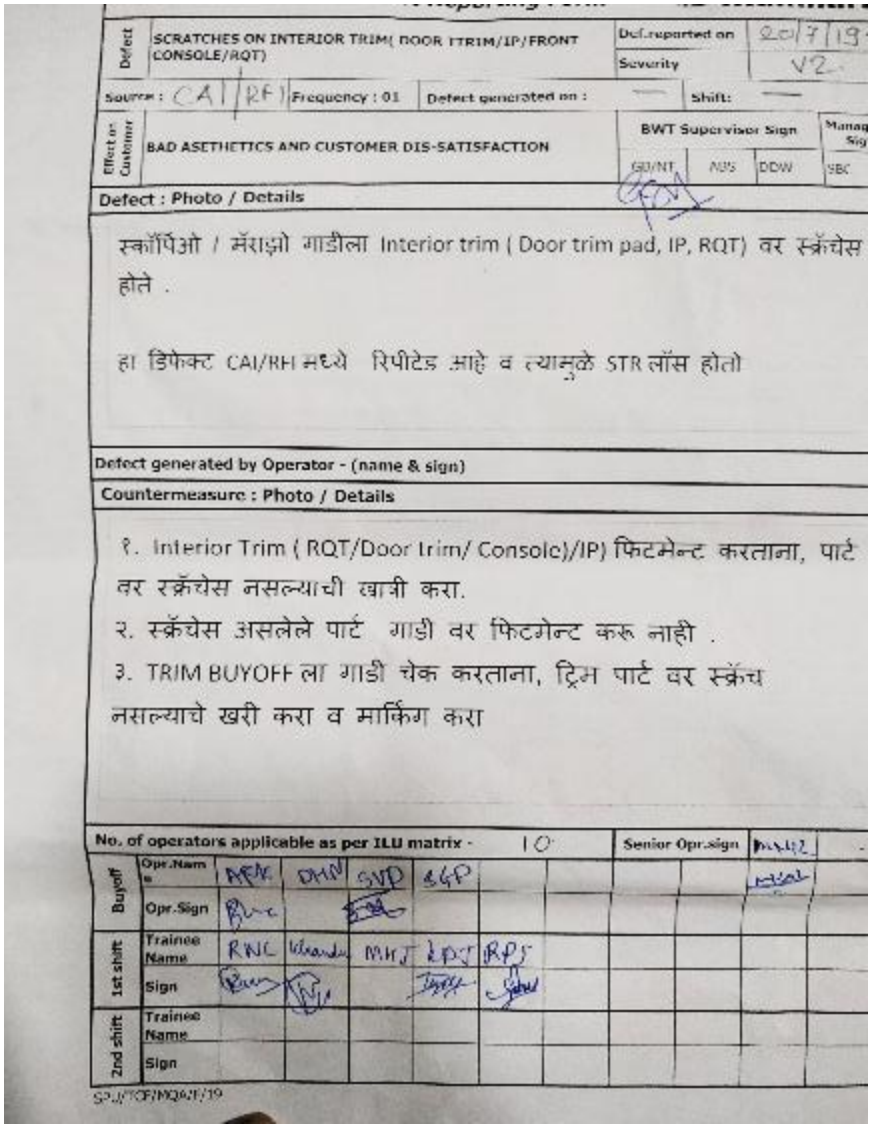
Incidences

Month

Inference : Average 43 No.s concerns reported per month from March -19 to May-19

2.b Immediate Containment Action

- Training for stage associates, explaining main step and key points.
- Emphasis on part handling and Tool handling while doing work.
- 100% In-coming lot checked for receiving scratches and damage
- Line side storage checked and condition restored by Foam application to immediately protect from scratches.
- DRF given to stage associates, explaining key points and focus on Dent/Damage/scratches.



Defect: SCRATCHES ON INTERIOR TRIM(DOOR TRIM/IP/FRONT CONSOLE/RQT)

Def.reported on: 20/7/19

Severity: V2

Source: CAI/RH | Frequency: 01 | Defect generated on: — | Shift: —

Effect on Customer: BAD ASETHETICS AND CUSTOMER DIS-SATISFACTION

BWT Supervisor Sign: GD/NT | NJS | DDW | SBC

Defect : Photo / Details

स्कॉर्पिओ / मॅराझो गाडीला Interior trim (Door trim pad, IP, RQT) वर स्क्रॅचेस होते .

हा डिफेक्ट CAI/RH मध्ये रिपीटेड आहे व त्यामुळे STR लॉस होतो

Defect generated by Operator - (name & sign)

Countermeasure : Photo / Details

१. Interior Trim (RQT/Door trim/ Console)/IP) फिटमेन्ट करताना, पार्ट वर स्क्रॅचेस नसल्याची खात्री करा.

२. स्क्रॅचेस असलेले पार्ट गाडी वर फिटमेन्ट करू नाही .

३. TRIM BUYOFF ला गाडी चेक करताना, ट्रिम पार्ट वर स्क्रॅच नसल्याचे खरी करा व मार्किंग करा

No. of operators applicable as per ILU matrix - 10 | Senior Opr.sign

Buyoff	Opr.Name	MPK	DHN	SVD	SGP					
	Opr.Sign									
1st shift	Trainee Name	RNC		MHJ	KPJ	RPJ				
	Sign									
2nd shift	Trainee Name									
	Sign									

SP-J/TOF/MQA/1/19

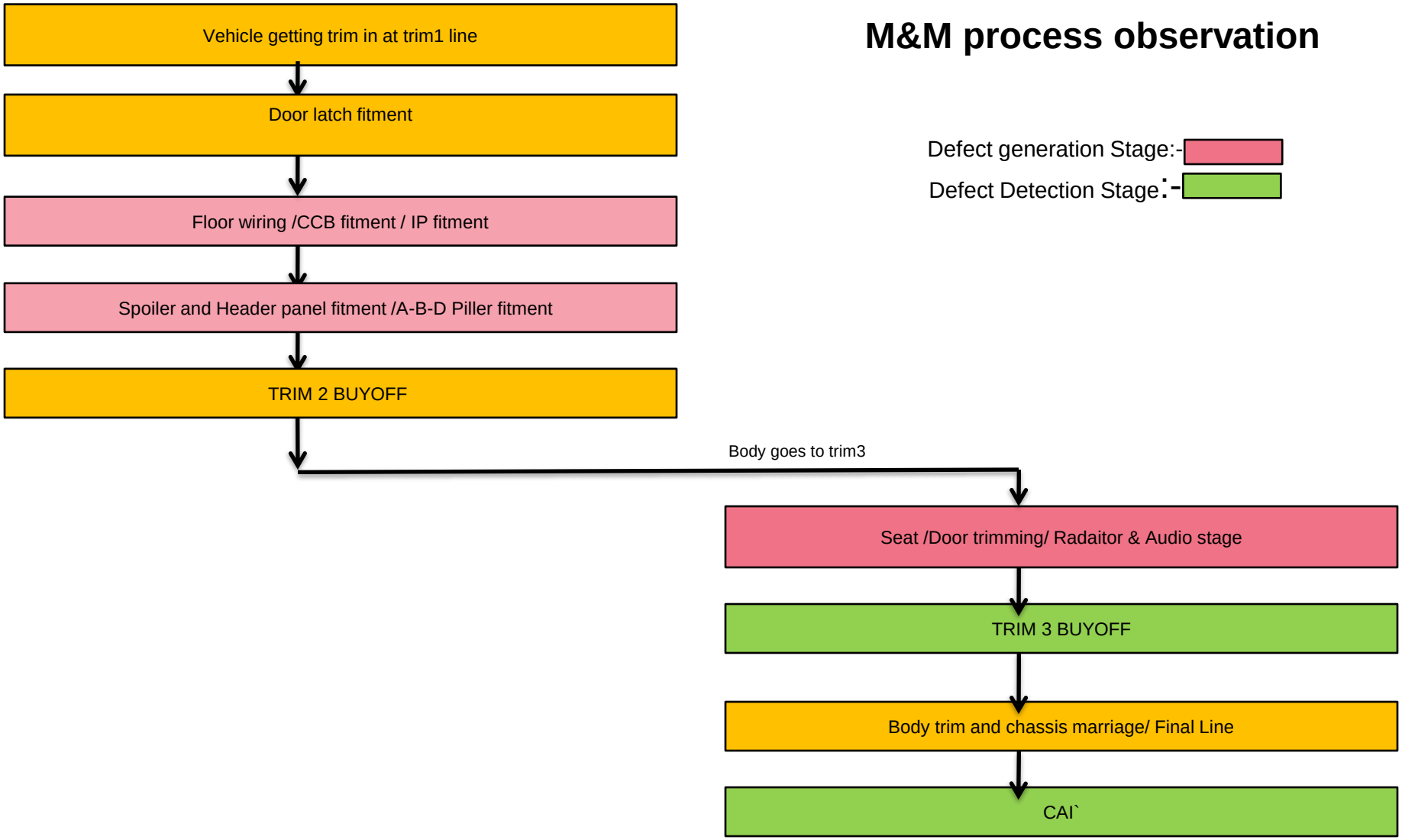
3. Understand the current situation

Flow Chart :- Observation of Process

M&M process observation

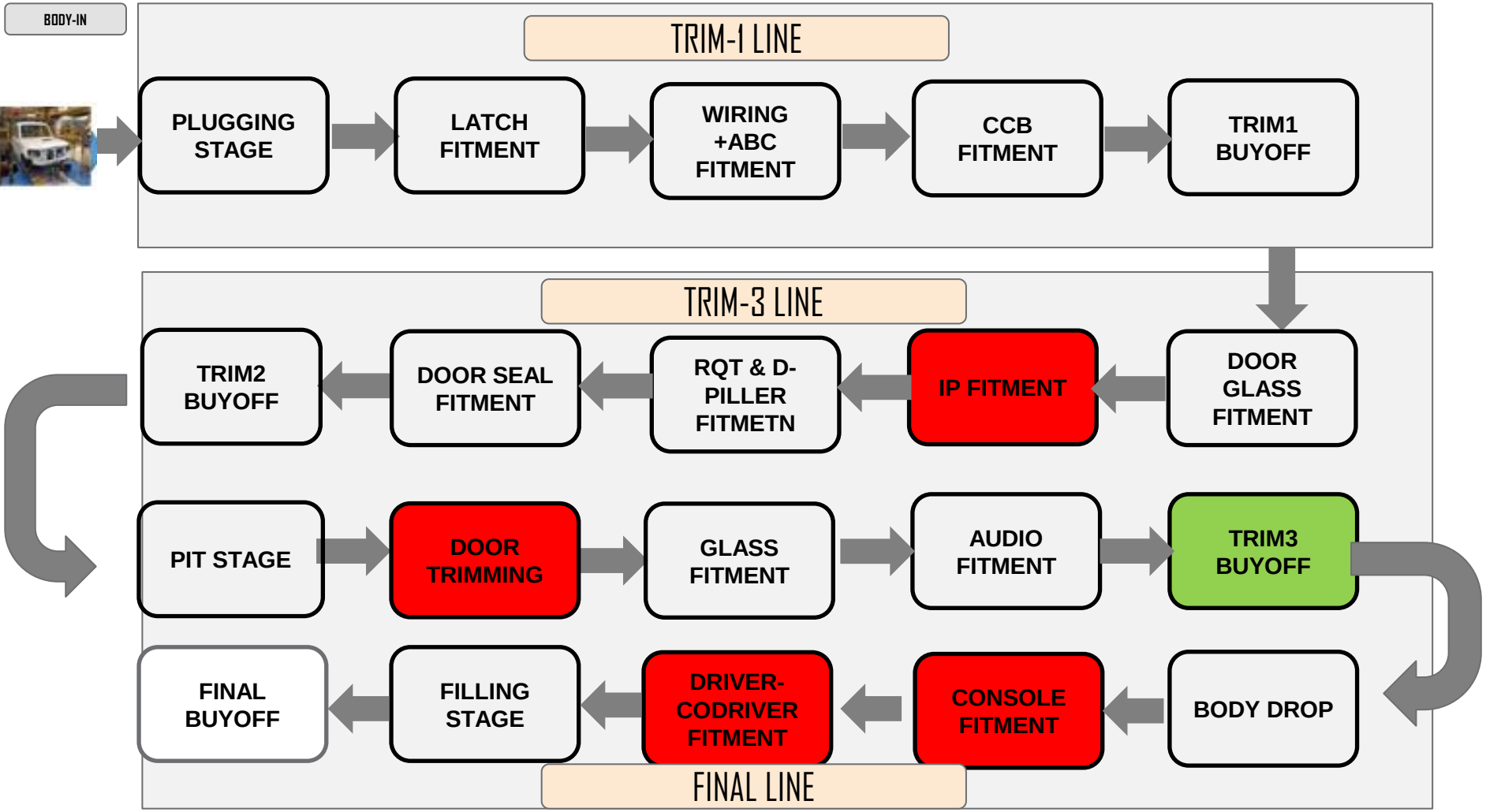
Defect generation Stage:-

Defect Detection Stage:-



Step 3. Understanding Current Situation

Observation of process



Defect generation stage
Defect identification stage

Step 3. Understanding Current Situation

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area :SCORPIO/XYLO TCF
Cell / BWT / Line: BWT-5/TRIM3

Requested By- MQA | Date -05.05.19 | Source -CAI/RFI | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : **SCRATCHES ON INTERIOR PARTS**

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 4 | VIN / Aggre. No.- | Model : SCORPIO/MARAZZO

Date of Incidence/ Assy : | Shift / Time : **BOTH SIFT** | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☐ CC ☑ SC

Immediate containment action for Customer Protection :
DRF GIVEN TO ALL STAGE ASSOCAITE
YARD CHECK DONE

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - GANESH BENDALE | BWT . : 5

Does a control already exist ? ☐ YES ☐ NO

Stage / Post Analysis -

Opearotor Name : **BHAGWAT V V (CAT:DR)**

Man(Human Action)
- Is operator trained ? ☑ YES ☐ NO
- Operator level (I.L.U) : I
- OR Dexterity level(1...4) : L3
- Is operator under training ? ☐ YES ☑ NO
- Is operator temporary for Stage ? ☑ YES ☐ NO
- Bad rework ? ☐ YES ☑ NO

Method(process)
- Main steps are respected ? ☑ YES ☐ NO
- Is there a related key point in SOP / SOS? ☑ YES ☐ NO
- Are the reasons for key point known ? ☑ YES ☐ NO
- Are operating reqmnts respected ? ☑ YES ☐ NO
- Is the defect part of the Buy off / Checkman list ? ☑ YES ☐ NO
- Was it detected by Inspector/ Buyoff Opr.? ☐ YES ☑ NO
- BUYOFF :SP DAPTARE

Other
- Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
- Is the packaging standard defined for the part ? ☑ YES ☐ NO
- Is the packaging standard respected ? ☑ YES ☐ NO
- Is there any alert mechanism for failure ? ☑ YES ☐ NO
- N/A

Machine
- Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
- Last calibration dates
- Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
- Is it a substitute facility ? ☐ YES ☑ NO
- Are machine parameters known ? (Record in Analysis table) ☐ YES ☐ NO
- Parameters are respected ? ☑ YES ☐ NO

Product Failure
- In only one shift ? ☐ YES ☑ NO
- In only one line ? ☑ YES ☐ NO
- The problem follow s the vehicle ? ☑ YES ☐ NO
- The problem follow s the part ? ☐ YES ☑ NO
- Is it a repeat problem ? ☑ YES ☐ NO
- Is the suspected parameter measurable ? (Record in Analysis table) N/A ☐ NO

Material/ Consumable
- Is the specified product used ? ☐ YES ☐ NO
- Did it evolve / Change ? ☐ YES ☐ NO
- Is the storage SOP respected ? ☐ YES ☐ NO
- Is the shelf life / expiry date respected ? ☐ YES ☐ NO
- Is the part as per FIFO? ☐ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -

1. Scratches observed on Interior Trims i.e., Door trim pad, Dash Board-IP, Front Console, RQT
2. At Gemba,it was found that while assembly and fitment,due to metallic part fouling to Interior trim,scratches is generated on Interior part.
3. Also Sequencing storage trolley found not ok- damage and wear out

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark
	NIL			

Not Applicable

Is 4M Analysis Part 2 required ? ☐ YES ☑ NO

Is the substitute transport mode used ? ☐ YES ☐ NO | **Is the packaging defined ?** ☐ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
.Awareness to stage and buyoff assocaite thorugh DRF	GANESH BENDALE	05.05.19	CLOSED

2. Permanent Action -	Resp	Date	Status
Modification in sequencing trolley			

Analysed by Name/s - GANESH BENDALE
Date - 05.05.19

Choice : ☑ WSS ☐ QC Story

L0 ANALYSIS FOR SCRATCHES ON INTERIOR PARTS

Inference from L0
Problem follows part & vehicle
Problem is a repeat problem
L1 analysis required

4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows part and vehicle
Material	Handling and shelf life of part adhered

NO QUICK WIN

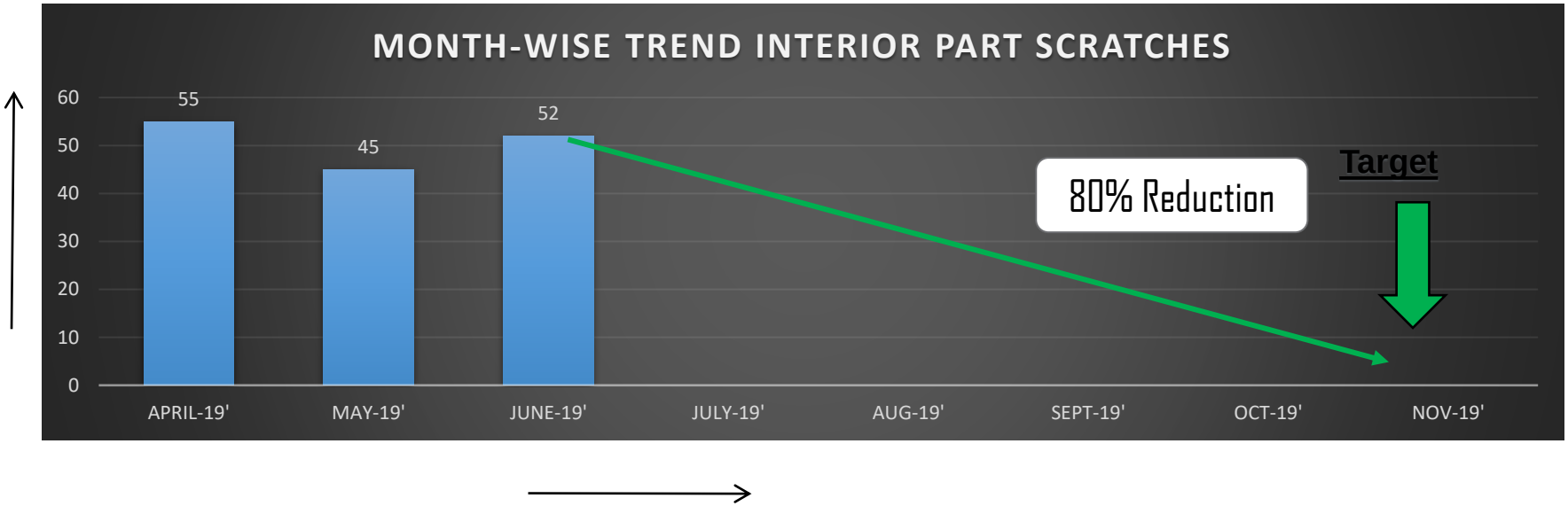
Step 3. Understanding Current Situation

Good & Not so good observations

- Issue observed only in Marazzo & Scorpio Model.
- Issue reported in both the shifts
- Repair time is high, around 8 minutes per vehicle
- Level 3 Associates are deployed for respective Interior trimming fitment.
- At IP and Door trimming stage-Lineside storage trolley rexine found wear out/damage.
- While Driver/Co-driver seat fitment-Possibilties of Sharp edges of seat bracket touching to interior part resulting into scratches on part .

4. Select targets

Present Level:	**Avg. 51 Incidences are being reported at CAI/BUYOFF.**
Target Level:	**Elimination of concern Interior part scratches by end of Nov - 19.**



Step 5. Analysis

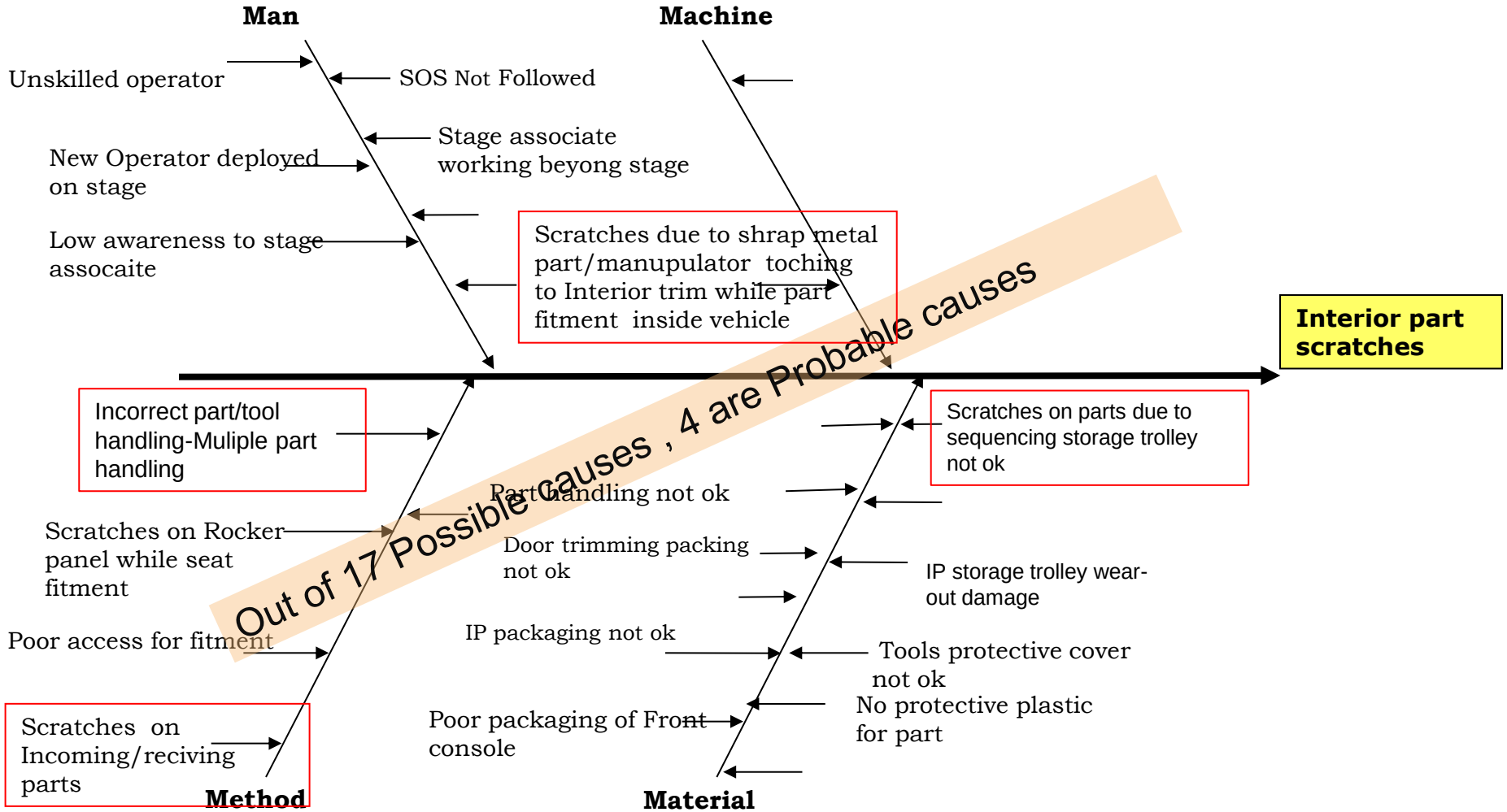
List of Possible causes identified in Brain storming session

Brainstorming Done along with CFT & we have found out 17 Possible Causes for this defect.

S N	Probable Cause
1	Unskilled operator
2	New Operator deployed on stage
3	SOS Not Followed
4	Low awareness to stage associate
5	Stage associate working beyond stage
6	Scratches on Incoming/receiving parts
7	Poor access for fitment
8	Scratches on parts due to sequencing storage trolley not ok
9	Scratches on Rocker panel while seat fitment by manupulator
10	Part handling not ok
11	Door trim packaging not ok
12	IP packaging not ok
13	Incorrect part/tool handling-Muliple part handling
14	Poor packaging of Front console
15	Scratches due to shrap metal part/manupulator touching to Interior trim while part fitment inside vehicle
16	Tools protective cover not ok
17	No protective plastic for part

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Probable causes :

S N	Probable Cause	Ganesh	Abhishek	Santosh	Satish	Nandy	SS Shelke	Total
1	Unskilled operator	**3**	**1**	**3**	**0**	**1**	**1**	**9**
2	New Operator deployed on stage	**3**	**3**	**3**	**3**	**3**	**1**	**16**
3	SOS Not Followed	**0**	**1**	**0**	**0**	**1**	**3**	**5**
4	Low awareness to stage assocaite	**1**	**0**	**3**	**1**	**3**	**0**	**8**
5	Stage assocaite working beyond stage	**0**	**0**	**0**	**1**	**0**	**1**	**2**
6	Scratches on Incoming/reciving parts	**9**	**3**	**9**	**3**	**3**	**9**	36
7	Poor access for fitment	**0**	**0**	**0**	**1**	**0**	**1**	**2**
8	Scratches on parts due to sequencing storage trolley not ok	**9**	**3**	**3**	**9**	**3**	**9**	**36**
9	Scratches on Rocker panel while seat fitment by manupulator	**1**	**3**	**0**	**1**	**1**	**0**	**6**
10	Part handling not ok	**0**	**3**	**1**	**3**	**0**	**3**	**10**
11	Door trim packaging not ok	**3**	**3**	**0**	**3**	**3**	**3**	**15**
12	IP packaging not ok	**3**	**3**	**3**	**1**	**1**	**3**	**14**
13	Incorrect part/tool handling-Muliple part handling	**9**	**3**	**3**	**9**	**3**	**9**	**36**
14	Poor packagingof Front console	**0**	**1**	**0**	**3**	**1**	**3**	**8**
15	Scratches due to shrap metal part/manupulator toching to Interior trim while part fitment inside vehicle	**9**	**9**	**9**	**9**	**9**	**9**	**54**
16	Tools protective cover not ok	**0**	**1**	**0**	**3**	**1**	**3**	**8**
17	No protective plastic for part	**3**	**1**	**0**	**3**	**1**	**3**	**11**

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 36.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering.

S N	Not Considered cause	Reason
1	Unskilled operator	**All Trained assocaites**
2	New Operator deployed on stage	**All Trained assocaites**
3	SOS Not Followed	**SOS are followed**
4	Low awareness to stage assocaite	**Found ok**
5	Stage assocaite working beyond stage	**Visually checked_Found ok**
6	Poor access for fitment	**No issue observed-30veh checked**
7	Scratches on Rocker panel while seat fitment by manupulator	**30 veh checked found ok**
8	Part handling not ok	**Lineside stock checked-found ok**
9	Door trim packaging not ok	**Visually checked_Found ok**
10	IP packaging not ok	**Visually checked_Found ok**
11	Poor packagingof Front console	**Visually checked_Found ok**
12	Tools protective cover not ok	**All Tools checked _Found ok**
13	No protective plastic for part	**Visually checked_Found ok**

5.1 Identify Probable cause

S N	Probable Cause
1	Scratches on Incoming/reciving parts
2	Scratches on parts due to sequencing storage trolley not ok
3	Incorrect part/tool handling-Multiple part handling
4	Scratches due to sharp metal part/manipulator touching to Interior trim while part fitment inside vehicle

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Scratches on Incoming/reciving parts	Visual check	Receiving and storage area stock checked- Door trim packaging found ok	Hypothesis Invalid

DOOR TRIMMING STORAGE AREA AUDIT :

As shown, Door trimming is coming in plastic wrap completely.

Following are the observation for Receiving and Storage Area checked for Door trim :-

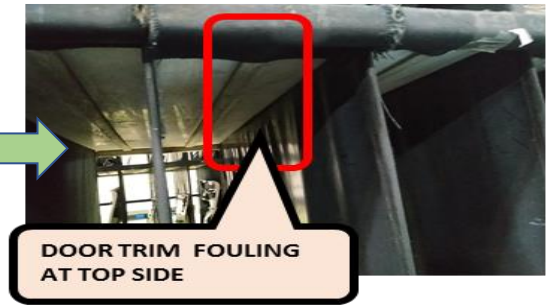
1. Door trim pad is coming in set (LH/RH).
2. Door trim are completely cover by Plastic by supplier-side.
3. PU Separator are provided for Door trim to avoid rubbing of Door trim and to avoid scratches

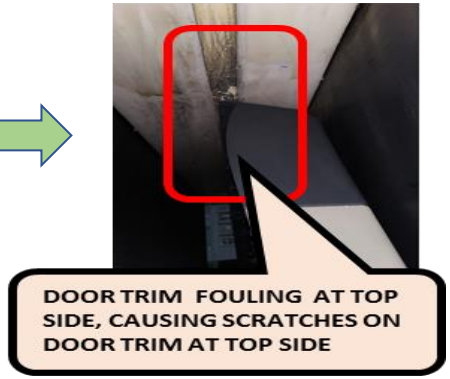
Conclusion :
As per above observation, Door trim packaging is in good condition. Hypothesis is InValid.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Scratches on parts due to sequencing storage trolley not ok	Visual check	Following are the observations for Sequencing Trolley part stoarage :- 1.Door trimming sequencing trolley found damage/wear out and Metal to Metal contact. 2.IP sequencing trolley found not ok-Sharp edges of trolley fouling to IP while unloading at stage.	**Hypothesis valid**





Conclusion –
Sequencing trolley condition found not ok-Rexine wear out and damage.
This resuts into scratches on trim parts

TEST OF HYPOTHESIS

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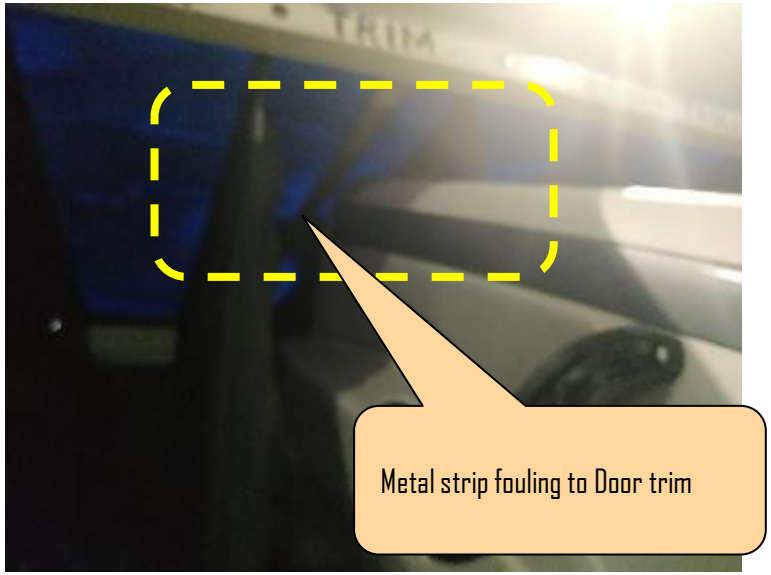
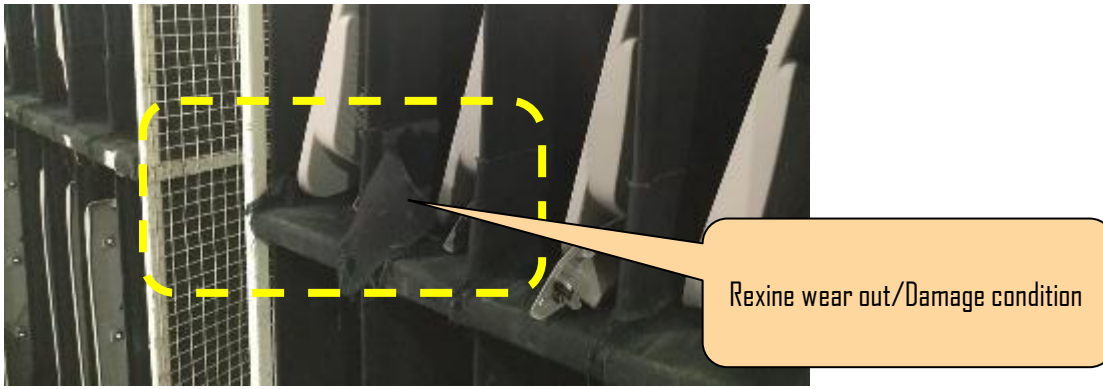
No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Scratches on parts due to sequencing storage trolley not ok	Visual check	Following are the observations for Sequencing Trolley part stoarage :- 1.Door trimming sequencing trolley found damage/wear out and Metal to Metal contact. 2.IP sequencing trolley found not ok-Sharp edges of trolley fouling to IP while unloading at stage.	**Hypothesis valid**



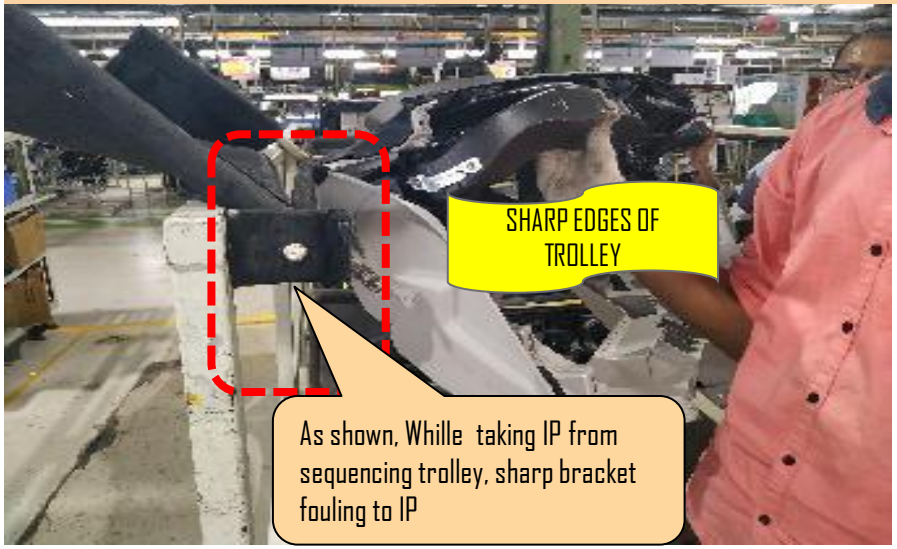


TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Scratches on parts due to sequencing storage trolley not ok	Visual check	Following are the observations for Sequencing Trolley part stoarage :- 1.Door trimming sequencing trolley found damage/wear out and Metal to Metal contact. 2.IP sequencing trolley found not ok-Sharp edges of trolley fouling to IP while unloading at stage.	**Hypothesis valid**

IP SEQUQNECING TROLLEY STORAGE AREA AUDIT :



Conclusion –
Sequencing trolley condition found not ok-Rexine wear out and damage. This resuts into scratches on trim parts

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Incorrect part/tool handling-Multiple part handling	Visual check	It is found that due to incorrect handling and tool handling by stage associate. Sharp pointed bit is causing scratches on trim parts	Hypothesis Valid

Incorrect **part handling** by stage associate.

Conclusion :
At Gemba, due to improper part handling/Tool handling result into scratches on part.
Hypothesis is Valid.

Incorrect **Tool handling** by stage assocaite

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Scratches due to sharp metal part/manipulator touching to Interior trim while part fitment inside vehicle	Visual check	Following are the observations at Final line which are causing scratches on Below interior parts due fouling of metallic part on Trim parts:- 1. Front console 2. Bothside Door trim parts	Hypothesis valid

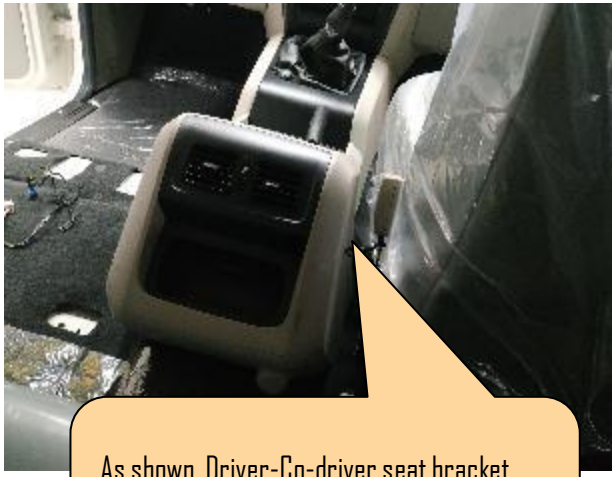
As shown, Driver-Co-driver seat bracket fouling to Door trim, resulting into deep scratches on Door trim

1. While Driver/Co-driver seat installation inside vehicle, seat bracket fouling to Door trim.
2. Sharp Bracket fouling to Door trim.
3. This results in to Deep Scratch on Door trim.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Scratches due to sharp metal part/manipulator touching to Interior trim while part fitment inside vehicle	Visual check	Following are the observations at Final line which are causing scratches on Below interior parts due fouling of metallic part on Trim parts:- 1. Front console 2. Bothside Door trim parts	Hypothesis valid



As shown, Driver-Co-driver seat bracket fouling to Front Console , resulting into deep scratches on Front Console

1. While Driver/Co-driver seat installation inside vehicle, seat bracket fouling to Front Console.
2. Sharp Bracket fouling to Front console .
3. This results in to Deep Scratch on Console (As shown)

Conclusion –
Stage associate followed the SOS correctly & ensures correct fitment process.
Till 4 Vehicles reported at CAI/Buy off for Scratches /Damage,due to seat bracket fouling to Inerior trim

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Scratches due to sharp metal part/manipulator touching to Interior trim while part fitment inside vehicle	Visual check	Following are the observations at Final line which are causing scratches on Below interior parts due fouling of metallic part on Trim parts:- 1. Front console 2. Bothside Door trim parts	Hypothesis valid

As shown,while IP fitment inside vehicle sharp edges of door flange fouling to IP.

1. While IP inside vehicle, seat bracket fouling to Front Console.
2. Sharp Edges of Door flange fouling to IP.
3. This results in to Deep Scratch on IP(As shown)

Conclusion –
Stage associate followed the SOS correctly & ensures correct fitment process.
Till 4 Vehicles reported at CAI/Buy off for Scratches /Damage,due to seat bracket fouling to Inerior trim

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Cause	Testing method	Conclusion
1	Scratches on Incoming/reciving parts	Visual Check	Hypothesis Invalid
2	Scratches on parts due to sequencing storage trolley not ok	Visual check	Hypothesis valid
3	Incorrect part/tool handling-Multiple part handling	Visual Check	Hypothesis valid
4	Scratches due to sharp metal part/manipulator touching to Interior trim while part fitment inside vehicle	Visual check	Hypothesis valid

3 Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION / COUNTER MEASURE
1	Scratches on parts due to sequencing storage trolley	Storage material trolley not in good condition	Storage material trolley rexine wear out/damage	Rexine wear / damage during handling	Trolley condition not maintained correctly	Correct the trolley
2	Incorrect part/tool handling-Multiple part handling	Part / tool fouling during other part fitment	Incorrect handling by associates	Lack of awareness about the severity od defect	Less focus on scratches / damage issues	Improve awareness
3	Scratches during part fitment assembly	Sharp metal part touching to trim parts	During assembly unknowingly, metal parts hit to trim parts	No protective cover to trim parts / trolley		Protective cover to be provide

6. Implement the corrective measures

Legends

↑ High ↓ Low

Possible Remedies Identified

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Trolley condition not maintained correctly	Correct the trolley	Low ↓	Low ↓	Yes	Low ↓	**Accepted**
2	Less focus on scratches / damage issues	Improve awareness	High ↑	Low ↓	No	Low ↓	**Accepted**
3	No protective cover to trim parts / trolley	Provide protective cover	Low ↓	Low ↓	Yes	Low ↓	**Accepted**

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Storage material trolley rexine wear out/damage	Correction in Sequencing trolley	10.09.19	Ganesh Bendale	Closed

Gemba Kaizen Sheet

Plant : Nashik	PU / Shop / Dept. : S.X TCF	Cell / Circle : BWT4
Machine / Equipment / Workstation No. :	Machine / Equipment / Workstation Name : SCUFPLATE	Date : 27.8.19

Pillar Applicability (pl. tick) ↓

KK / TM ☐	JH ☐	PM ☐	QM ☐	DM ☐	E&T ☐	OTPM ☐	SHE ☐

Theme : ELIMINATION OF DOOR TRIM SCRATCHES

BEFORE

AFTER

DOOR SEQUNECING TROLLEY MODIFIED-ACKYLRIC SEPARATOR REPLACED BY REXINE AND STORAGE WIDTH INCREASED TO AVOID RUBBING OF DOOR TRIM CAUSING SCRATCHES (TOTAL DOOR TRIM TROLLEY MODIFIED=06)

Description in details :-	Description in details :-
SRATCHES/DENT/DAMAGE ON DOOR TRIM (TOP CONCERN AT TRIM3 BUYOFF AND CAI)	DOOR TRIM TROLLEY MODIFIED-ACKYLRIC SEPARATOR REPLACED BY REXINE AND STORAGE WIDTH INCREASED TO AVOID RUBBING OF DOOR TRIM CAUSING SCRATCHES

KEY BENEFITS

Sr.no.	Description	Benefits
1	Elimination of Scratches	CAI STR improved
2		

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Storage material trolley rexine wear out/damage	Correction in Sequencing trolley	10.09.19	Ganesh Bendale	Closed

SEQ. TROLLEY NO	SEQUENCING TROLLEY HEALTH			REMARKS
	DAMAGE	REXINE TORN	METALLIC PART OPEN	
1	✓	✓	✓	OK
2	✓	✓	✓	OK
3	✓	✓	✓	OK
4	✓	✓	✓	OK
5	✓	✓	✓	OK
6	✓	✓	✓	OK
7	✓	✓	✓	OK
8	✓	✓	✓	OK

6. Implement the corrective measures

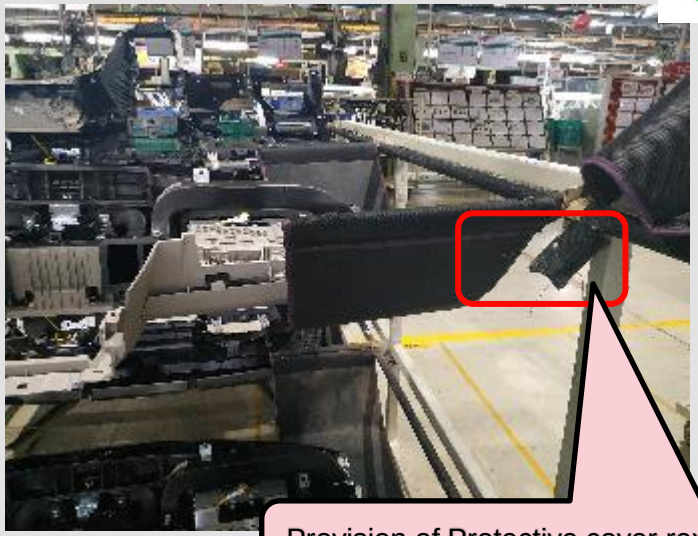
6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Storage material trolley rexine wear out/damage	Correction in Sequencing trolley	10.09.19	Ganesh Bendale	Closed

BEFORE

Due to improper storage, switch lugs get damage/bend. (As shown Fig1 and Fig. 2)

AFTER



Provision of Protective cover rexine on Bracket

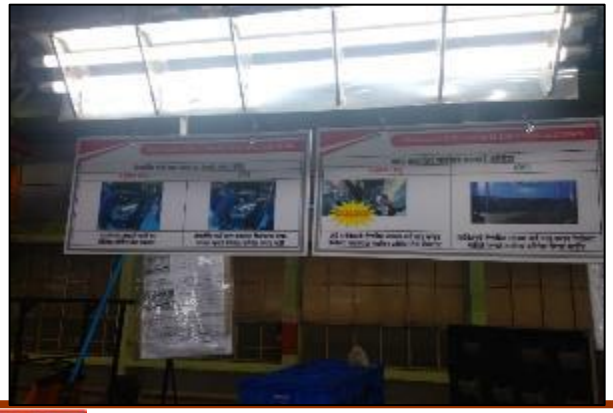
6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
2	Stage associate not aware severity for scratches/Damages	Cultural improvement to all stage and Buyoff Buyoff associate,giving emphasis on Scratches /Dent/Damage	21.08.19	Ganesh Bendale	Closed

CONT'D...

DO's AND DON'T BOARD DISPLAYED AT LINE SIDE



Slogan Boards Displayed on Vehicles for Awareness to all assocaites

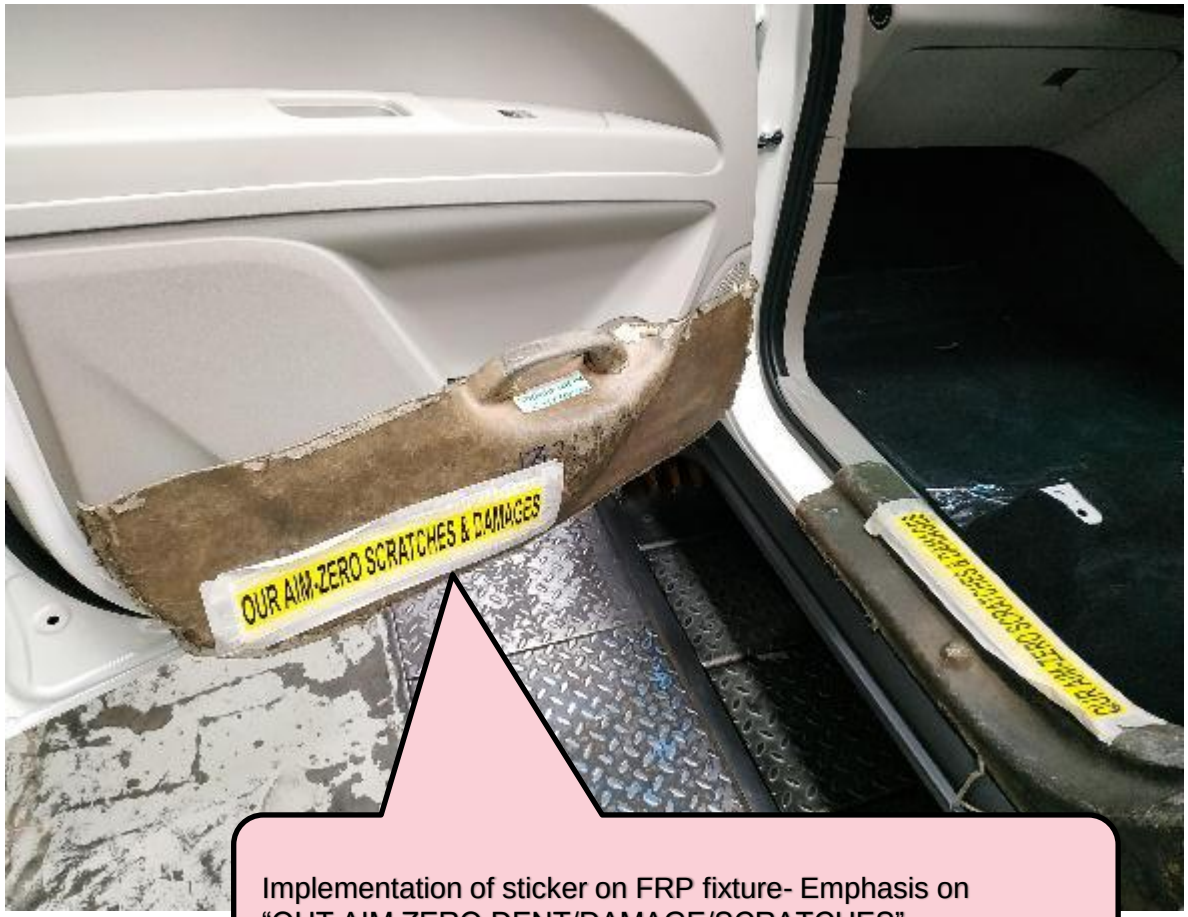
SLOGAN BOARD DIPLAYED ON VEHICLE, CREATING AWARENESS FOR ZERO DENT/DAMAGE/SCRATCHES

Slogan Boards Displayed on Vehicles for Awareness among All

Associate Awareness 21st July 2019

❑ Vehicle Dent, Damage, Scratches Awareness

IMPLEMENTATION OF STICKER ON FRP FIXTURE



Implementation of sticker on FRP fixture- Emphasis on
“OUT AIM ZERO DENT/DAMAGE/SCRATCHES”

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
3	No protective cover to trim parts	Provision of FRP Fixture	**10.09.19**	Sachin Bhamare	CLOSED

BEFORE

While seat Fitment seat bracket fouling on console resulting scratch genrate on console

NO PROTECTIVE COVER ON FRONT CONSOLE

AFTER

PROVISION OF PROTECTIVE FRP FIXTURE ON FRONT CONSOLE

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
3	No protective cover to trim parts	Provision of FRP Fixture on Door trim	**15.09.19**	AMIT TILE	CLOSED

BEFORE

While seat Fitment seat bracket fouling to Door trim results into deep scratch on Door trim

NO PROTECTIVE COVER ON FRONT CONSOLE

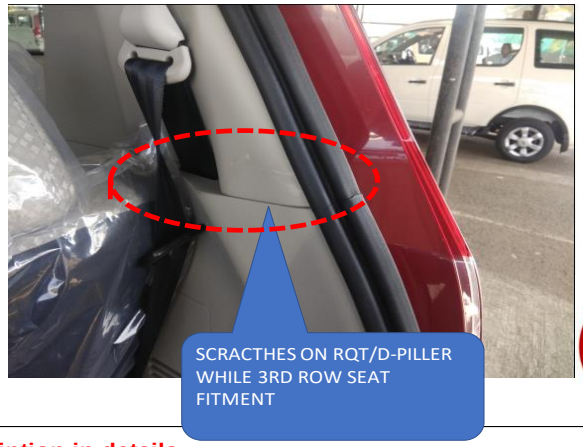
AFTER

PROVISION OF PROTECTIVE FRP FIXTURE ON FRONT CONSOLE

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
3	No protective cover to trim parts	Provision of FRP fixture	**05.09.19**	Sachin Bhamare	CLOSED

BEFORE	AFTER
	
Description in details :-	**Description in details :-**
Scratches on RQT (Marazzo Veh) due to seat bracket touching to RQT at seat stage	**Provision of FRP to avoid scratches and damage to veh. Cut off no. 17991**

KEY BENEFITS

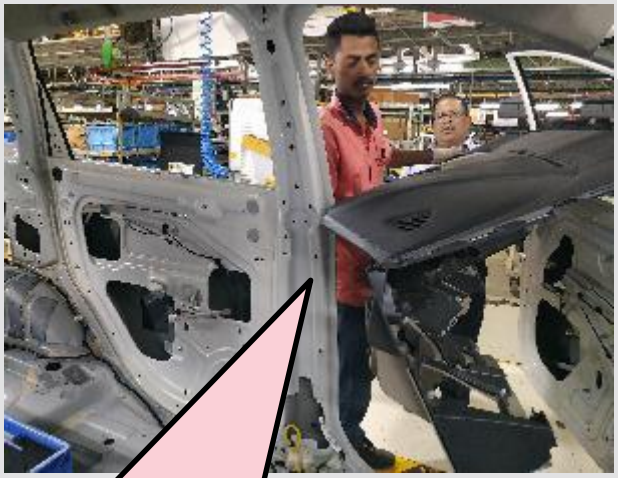
Sr.no.	Description	Benefits
1	Elimination of Scratches & damage	CAI STR improved
2		

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
3	No protective cover to trim parts	Provision of FRP fixture	**05.09.19**	Sachin Bhamare	CLOSED

BEFORE



While IP fitment inside vehicle,
IP fouling to sharp door flange

AFTER

Implementation of sticker on FRP fixture for IP fitment

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
3	No protective cover to trim parts	Provide protective cover	**05.08.19**	Ganesh Bendale	CLOSED

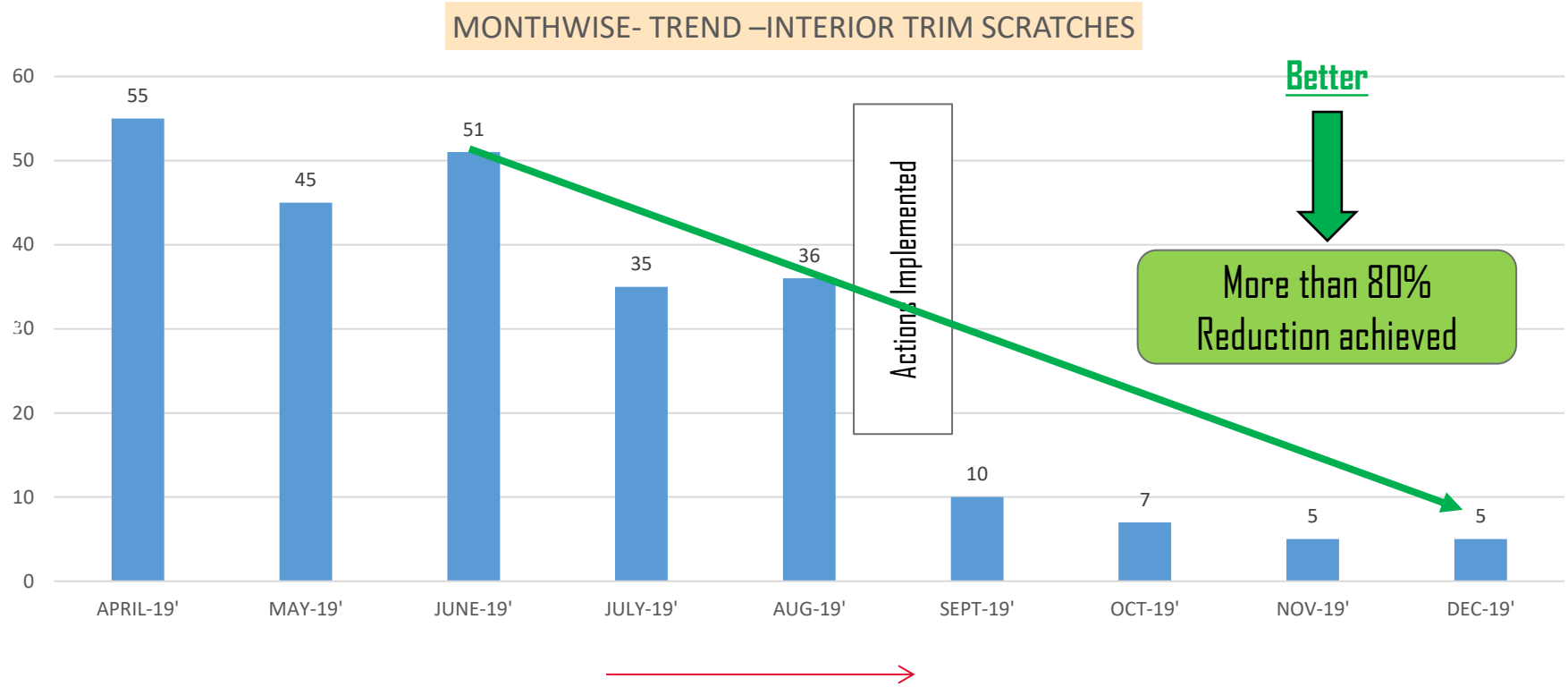
BEFORE	AFTER
Description in details :-	**Description in details :-**
Door stage trolley wear out and in damage condition, sharp edge causes major scratches on veh	**Provision of rexine and foam to door stage trolley to avoid scratches and damage to veh**

KEY BENEFITS

Sr.no.	Description	Benefits
1	Elimination of Scratches & damage	CAI STR improved
2		

7. Check & confirm the results & effects

7.1 Verify Results:



Inference – Concern not reported from after implementing the PCA actions.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S.N	Action Taken	Actual date	Responsibility	Status
1	Correction in Sequencing material storage trolley	10.09.19	Ganesh Bendale	CLOSED
2	Cultural improvement to all stage and Buyoff Buyoff associate,giving emphasis on Scratches /Dent/Damage	21.08.19	Ganesh Bendale	Closed
3	Provision of FRP Fixture	15.09.19	Sachin Bhamare	CLOSED
4	Work-Content Element sign off	15.09.19	Ganesh Bendale	Closed
5	PFMEA updation (Rev No –02)	20.09.19	Ganesh Bendale	CLOSED
6	Process audit check sheet updation (Rev No -6)	20.09.19	Sachin Bhamare	CLOSED
7	SOS Updation (Rev No - 02)	25.09.19	Ganesh Bendale	CLOSED
8	On job Training given to stage associates for power window switch bezel fitment.	25.07.19	Ganesh Bendale	Closed

Check & confirm the results & effects (Cont……)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of concern Scratches on Interior trim
2) Repeat V2 concerns of RFI/GNOVAC audit is eliminated completely.
3) DPT(FTBO) due to Scratches was **123** which is now reduced to **zero.**
4) Rework cost save upto 76412.9.94 Rs / Year

In tangible Benefits

1) Improved Customer Satisfaction.
2) Operator fatigue reduced , moral improved
3) Vehicle aesthetic improved.

DHANOKAR RAJESH | BENDALE GANESH; CHIKHALE SANDEEP; BHAMARE SACHIN

RE: Rework Cost estimation required for MYB project

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	4.0
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	254.71
Avg no. of working days	days	25
Operator cost /month	Rs	6367.7
Operator Cost/year	Rs	76412.9

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S.N	Action Taken	Actual date	Responsibility	Status
1	PFMEA updation (Rev No –02)	20.09.19	Rahul Gadekar	CLOSED
2	Process audit check sheet updation (Rev No -6)	20.09.19	Sachin Bhamare	CLOSED
3	SOS Updation (Rev No - 02)	25.09.19	Ganesh Bendale	CLOSED
4	PCRN	14.11.19	Sayali More	CLOSED
5	On job Training given to stage associates for power window switch bezel fitment.	25.07.19	Ganesh Bendale	Closed

8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	On job Training given to stage associates	25.07.19	Ganesh Bendale	Closed

8.3 Additional Audit & check :

SN	Document	Target date	Responsibility	Status
1.	Daily Process audit sheet updated	25.09.19	Ganesh Bendale	Closed

8. Standardization

8.1 Updation of documents :-

PFMEA Updation

Plant : Nashik	Model : Z	Mahindra & Mahindra Ltd.	Cross Functional Team
Part No/s.:	Part Na		Suhas Aher, Bbab Awhale
Latest Change Letter:	Optn No.: 05010, 4990	Process Potenial Failure Mode & Effect Analysis.	Vikrant Ghule, Santosh Jadhav
PU/ Shop :Scorpio TCF	Optn Despt:- Driver seat fitment	Reviewed by :Suhas Aher; Rev. No : 59; Doc No.: PFMEA/SPU/SCORPIO/FINAL/5030	Sayali More
Module/ Cell :Final	M/C Name & No.:	Approved by : Mandar Joshi; Rev. Date : 05.08.19; FMEA Date (Orig) :25/11/00	

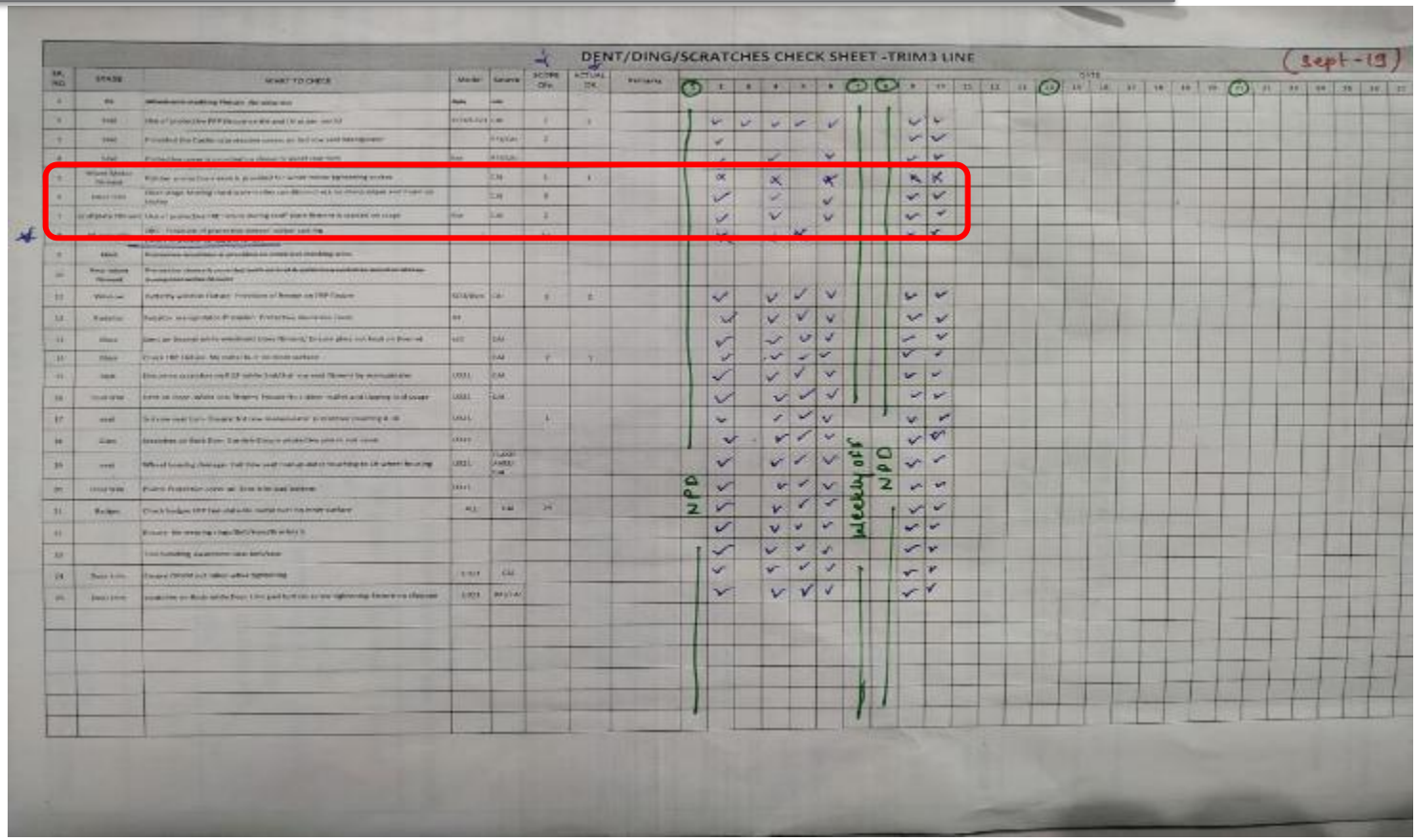
Process Function	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
	Latch wire clipping on seat.	Wire clipping not done.	**Mfg. & Assy:** Rework at post station. (5) **End user:** May trapped in seat.(4)	4		Operator forgot to clipped on seat.	Awareness givent to stage operator through drf	2	Check point at buy off.	7	56							
		Mounting clip broken	**Mfg. & Assy:** Rework at post station. (5) **End user:** May trapped in seat.(4)	4		Clip broken durring seat handling from Lear to M & M	Awareness givent to stage operator through drf	2	Check point at buy off.	7	56							
	Latch socket connection .(Buy Off Concern)	Connection not done.	**Mfg. & Assy:** Rework at post station. (5) **End user:** Safety norms voilating..(8)	8	sc	Operator forgot to connect the socket.	Awareness givent to stage operator through drf. Relative marking started on socket by stage operator itself.	6	Check point at rolled down stage and Intrim buy off'	7	336	Awareness to be given all associates	Rahul Gadekar	Action Completed				
	2) Collect driver seat from trolley as per model	Wrong type of seat fitted.	**Mfg. & Assy:** Rework at post station. (5) **End user:** Poor appearance.(4)	9	C C	Operator selects wrong seat	1) Identification sticker on seat & sequential feeding on line. 2) Difference according to the model is mentioned in SOP which is displayed on stag .	6	1) check at final buy-off. (7) 2) At CAI (8) 3)Seat Checking pokayoke at Cai	7	378							
	3) Place seat inside vehicle carefully .	Damage/scratches to body/Door Trimpad/ Floor console (parts fitted on body) while Placing .	**Mfg. & Assy:** Rework at post station. (5) **End user:** Poor appearance.(4)	3		Improper handling due to Heavy & bulky seats	Training to operator.	4	1) For seat handling manipulators provided 2) Check at CAI	7	84	Provision FRP Fixture for Door trim and Console	Rahul Gadekar	Action Completed	3	3	7	63
	4) Align holes on seat bracket & body ,Prefit bolts .	Bolt not fitted.	**Mfg. & Assy:** Rattle Noise, Unsafe condition and rework. (5)	5		Misalignment of seat	Locating lugs on rear seat mtg. Bracket	4	1)Cutout for seat mounting on floor melt sheet to be increased. 2) Self certification (5). 3) At CAI. (8)	7	140							
	2) Collect driver seat mtg bolt 2 type.	Wrong bolt fitted (Instead of small bolt big bolt fitted) Undre body noise.	**Mfg. & Assy:** Rework at post station. (5) **End user:** Under body noise.(5)	7		Operator fitted wrong bolt.	Awareness given to all stage operator through drf. Golden passivation done to bolt for easy identification	3	1) check at final buy-off. (7) 2) At CAI (8)	7	147							
	5) Tightening of seat mtg. Bolts.	1) Bolt half torque. **(Buy Off Top Concern)**	**Mfg. & Assy:** Rattle Noise & rework.(5) **End user:** Rattle noise & Unsafe condition (9).	9	C C	Operator cross fit the bolt	1) Self certification. 2) Training to operator. 3)DRF given to all stage operator.	6	1) Check point added at LH buyoff (Feeling with hand and visual checking). 2) Marking started at both buyoff with marker on all bolt. 3)SOS modified, checking the gap between bolt head and body floor started and like wise trained to operator	7	378	Daily process monitoring through process audits, IQ audits, S.A. checklist and superwiser checksheet.	Rahul Gadkar June 2019	Action Completed				
		2) Bolts under torqued / Loose.	**Mfg. & Assy:** Rattle Noise & rework.(5) **End user:** Rattle noise & Unsafe condition(9).	9	C C	Improper power tool calibration	Periodic Tool Calibration	6	Torque audit	7	378							
						Use of wrong power tool	Dedicated power tool .		Identification sticker on Power tool									
		3) Bolts overtorqued	**Mfg. & Assy:** Rattle Noise & rework.(5) **End user:** Rattle noise & Unsafe condition(9).	9	C C	Improper power tool calibration	Periodic Tool Calibration	2	Torque audit	7	126							
						Use of wrong power tool	Dedicated power tool .		Identification sticker on Power tool									

Rev No.	Rev Date	Revision Summary	Reason for Revision
49	28.11.2017	Driver seat bolt half torque- LH RR	Export RFI Concern
50	12.12.2017	Deep Scratches Trim Pad Front Door LhS	Export RFI Concern
51	21.02.2018	NO Change	Six Monthly Review
52	13.03.2018	Console scratches	RFI concern updated
53	09.03.2018	Review for EN - Front Driver seat buckle wiring harness length change from 330 to 450	EN NO. W105-EN-A003-000399
54	31.05.2018	Seat latch wire clip broken	G Nova C concern
55	07.08.2018	NO Change	Six Monthly Review
56	04.10.2018	Review for New Model introduce S9	EN change
57	12.02.2019	NO Change	Six Monthly Review
58	20.07.19	Implementation of FRP fixture for Door trimming	RFI/CAI concern
59	05.08.2019	Change in OCC	Six monthly review

8. Standardization

8.3 Additional Audit & check

Daily Process audit check sheet updation :-



8. Standardization

8.3 Additional Audit & check

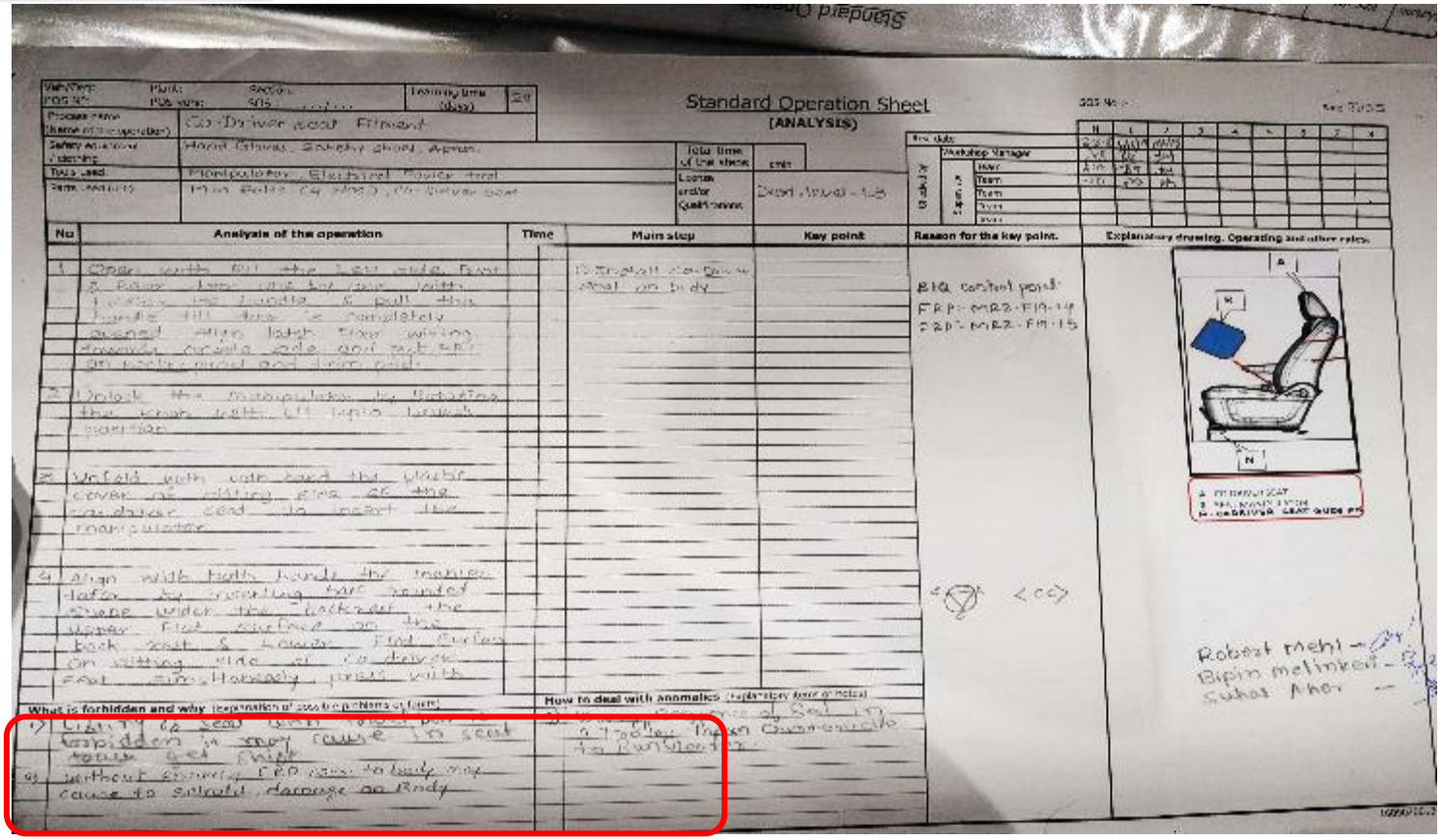
Daily Process audit check sheet updation :-

1	List of FTG Which are touching to Body (Final Line)											
2	SR. NO	FTG	WHAT TO CHECK	WORK STATION	Qty	Freq	Month-					
3							V1	V2	V3	V4	V5	
4	1	Coupling protective cover	Coupling Fully covered	All stage	86	Weekly						
5	2	Console Anti scratch cover RH / LH for scorpio	FRP fixture -No wear out from inside (Total=2)	seat stage	2	Weekly						
6	3	Trim pad anti scratch cover for S4+ ,S5, S7, S11 - Both side LH / RH	FRP fixture -No wear out from inside (Total=2)	seat stage	2	Weekly						
7	4	Front bumper puller fixture	Check welding for fixture	seat stage	1	Weekly						
8	5	Front wind shield glass FRP For marazzo	FRP fixture -No wear out from inside	Console/ battery	10	Weekly						
9	6	Fender cover (protective cover) FRP Scorpio	FRP fixture -No wear out from inside	Console/ battery	1	Weekly						
10	7	Fender cover (protective cover) FRP W103	FRP fixture -No wear out from inside	Console/ battery	1	Weekly						
11	8	Marazzo Head Lamp protective cover	FRP fixture -No wear out from inside	Console/ battery	1	Weekly						
12	9	Marazzo rocker panel FRP	FRP fixture -No wear out from inside	seat stage	2	Weekly						
13	10	Trim pad anti scratch cover for marazzo	FRP fixture -No wear out from inside (Total=2)	seat stage	2	Weekly						
14	11	Front bumper Trolley	All edges coverd, no sharp edges.	Console	3	Weekly						
15	12	Rear bumper Trolley	All edges coverd, no sharp edges.	Console	3	Weekly						
16	13	Articular arm - RHS	Balancer fully coverd	Body drop	1	Weekly						
17	14	Articular arm - LHS	Balancer fully coverd	Body drop	1	Weekly						
18	15	Tail gate 'D' cal Fixture - W103 SC/DC	FRP fixture -No wear out from inside	DL 10	1	Weekly						
19	16	Tail gate 'D' cal Fixture - W109 SC/DC	FRP fixture -No wear out from inside	DL 10	2	Weekly						
20	17	Tail gate trimming bin	Bin fully covered from bottom	Cargo trimming	22	Weekly						
21	18	FRP (Body protective) at LH / RH D pillar For SC/DC	FRP fixture -No wear out from inside	cargo drop	2	Weekly						
22	19	FRP for tail gate Badge fitment	FRP fixture -No wear out from inside	Cargo trimming	3	Weekly						
23	20	Battery negative cable setting tool & hammer	Tool not get damge from lower side	Filling	1	Weekly						
24	21	Diesel bleeding gun	junction box fully coverd	Diesel bleeding	1	Weekly						
25												

8. Standardization

8.1 Updation of documents :-

SOS Updation



8. Standardization

8.1 Updation of documents : -

PCRN:

CREATE PCRN

PCR...:* PCRN/A003/STCF/F20/14879 | Use of fxiture to avoid scratches | Planned closing date:* 31.12.2019 | Actual closing date:

A.Form

PU / Area:* SCORPIO TCF | Cell:* FAI | Date: 14.11.2019

Models Affected: All

Opn No./Mc No.: | Opn Description: Use of fxiture to avoid scratches

Process Change Trigger

- [ ] Process Parameters
- [ ] Layout Change
- [ ] Production from New M/c
- [ ] Poka Yoka Introduction
- [ ] Inspection & Test Method
- [ ] Reconditioned M/c/ Tooling
- [ ] Method
- [ ] Use of alternate material
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Transferred M/c
- [x] Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- [ ] Tools
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Change of Vendor for Part Material, Paint etc
- [ ] Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified

Use of fxiture to avoid scratches on INTERIOR TRIMS I.E DOOR TRIM, FRONT CONSOLE,ROCKER PANEL

Estimated Cost Impact

Estimated Cost Impact: ○ Saving ○ Increase ◉ Others

- [ ] Material
- [ ] Manpower
- [ ] Conversion

E.Requesting PU/ Area / Shop

1Does this Change affect other operations in the PU If yes, list the affected operations: ○ Yes ◉ No

2.Are any other activities/operations getting added to other PU's /Other LocationsVendors ○ Yes ◉ No

i.List the affected operations | ii.Affected PU/Location /Vendor

Benefits or Impact from PCRN

Applicability	Description	Present	Expected	Applicability	Description	Present	Expected
☐	Rework / Rejection/PPM :			☐	Cost of Operation / Process :		
☐	Line Throw-back :			☐	Operation cycle time :		
☐	Warranty Failure :			☐	Tool Set-up time :		
☐	Improvement in Cp / Cpk :			☐	Machine Down time :		
☐	Tooling Cost :			☐	Ergonomic Study Status of stage/opn (GYR status):		
☐	Line TACT time/Work Content :			☑	Any Other Parameter:		

Upload Documents(Attachments limit is 1mb/PCRN no.)

A.

Approval Status

8. Standardization

8.1 Updation of documents : -

QA MATRIX:

Section Manager	1	2	3	4	Revision date	12.11.19

Workstation No.	Process description (POS)	Date	Failure modes	Severity	Prev. occurence	Potential decfect	Inspection process – Section: PIT	SEAT	DOOR	BMT6 BUYOFF	FAI	Outside section: Rolldown	Roller	tester	Shower	S&R	ECOS	WRAP	Guarantee value: Workstation	Section	Outside section	PLANT	Faults recorded: Section (1M)	Outside section (1M)	CUSTOMER (3M)	Quality guarantee level: WORKSTATION	Section	Outside section	PLANT
	Waist seal RH RR door inner	26.12.19	Misassembled of	V2 3					1	1									1	2	0	2	0	0	0	NOK	NOK	NOK	NOK
	Door trim pad	12.11.19	Scratches on Door trim	V2 3					1	3	1							1	1	4	2	6	0	0	0	OK	OK	NOK	OK

8. Standardization

8.1 Updation of documents : -

ELEMENT SIGNOFF:

MOST Element Report

Operation Group : SCORPIO TCF F20
Operation No : 10
DOOR INNER FRP COVER FITMENT & REMOVAL BEFORE IP LOADING LH SIDE_MARAZZO_IP STAGE

SR NO.	ELEMENTS DESCRIPTION	FRQ.	DIV	MEN	OFFLINE
1	MOVE 3-4 + GRASP FRP + BACK + PLACE ON DOOR INNER ABOVE DOOR STRIKER	1.00	1.00	1.00	No
1	MOVE 3-4 STEPS, GRASP OBJECT , MOVE 3-4 STEPS, PLACE WITH ADJUSTMENT	1.00	1.00	1.00	NO
2	PRESS PROPERLY	2.00	1.00	1.00	No
1	PUSH PULL SWITCH / ROTATE KNOB, SLIDE / PUSH BUTTON / PUSH/PULL/LEVER/PIVOT <=12"	1.00	1.00	1.00	NO
3	*********REMOVAL AFTER WORK	1.00	1.00	1.00	No
4	MOVE 3-4 + GRASP FRP FROM VEHICLE + BACK + PUT ASIDE	1.00	1.00	1.00	No
1	MOVE 3-4 STEPS, GRASP OBJECT , MOVE 3-4 STEPS, PUT	1.00	1.00	1.00	NO

9. Summarize & Plan future actions

9.1 Conclusion:

Scratches on Interior trim Concern eliminated Provision of FRP fixture for Interior trim, Cultural improvement among all stage assocaites

9.2 Horizontal Deployment:

Horizontal Deployment for Upcoming New projects

9.3 Lessons Learnt:

Things Gone Wrong : Failure mode of seat bracket fouling to Trim parts not identified at time implementation.

Things Gone Right : Understood the importance of Provision of FRP fixtures and its mandatory consideration in process.

Thank you

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